Sunghyun Woo

wshyun314@gmail.com | Google Scholar | LinkedIn | GitHub

Technical Skills

Programming: C++, Python, CUDA
Frameworks & Tools: PyTorch, OpenCV, Git, Docker, Conda
Research Areas: 3D Reconstruction, Novel View Synthesis, 3D Vision, Computer Vision, Deep Learning

Experience

Research Intern — Apr. 2025 – Present
Media for Machine Lab (M4ML), Dong-A University (Advisor: Prof. Jeongil Seo) — *Busan, South Korea*

- Conduct research on neural scene representations and 3D reconstruction, with a focus on Neural Radiance Fields and 3D Gaussian Splatting.

Publications

Densification Rescheduling for 3D Gaussian Splatting Performance Improvement — 2026

- **Sunghyun Woo**, Jeongil Seo*
  Proceedings of the Korean Institute of Broadcast and Media Engineers Conference, pp. 1047–1050 — paper

Comparative Study on the Performance of NeRF and 3DGS Based Models — 2025

- **Sunghyun Woo**, Jeongil Seo*
  Journal of Broadcast Engineering, vol. 30, no. 6, pp. 927–941 — paper

Quality-Enhanced 3D Gaussian Splatting via Tile Hard Mining and Edge Weighted Densify — 2025

- **Sunghyun Woo**, Jeongil Seo*
  Proceedings of the Korean Institute of Broadcast and Media Engineers Conference, pp. 24–27 — paper

Neural Radiance Fields and 3D Gaussian Splatting: Performance Comparison Analysis by Iteration — 2025

- **Sunghyun Woo**, Jeongil Seo*
  Proceedings of the Korean Institute of Broadcast and Media Engineers Conference, pp. 1530–1533 — paper

* Corresponding author.

Honors & Awards

2026 – Excellence Award, One Punch Ralph-thon in Busan
2025 – Excellence Award, Capstone Design Competition (Poster Presentation), Korean Institute of Broadcast and Media Engineers Summer Conference
2025 – First Prize, SZU–DAU International Hackathon

Education

Dong-A University — Busan, Korea
B.S. in Department of Computer Engineering; GPA: 4.13 / 4.5 — *Mar. 2022 – Present*